SDA SD Express Student Challenge

"EmberEye"

System for Real-Time Detection of Wildfires

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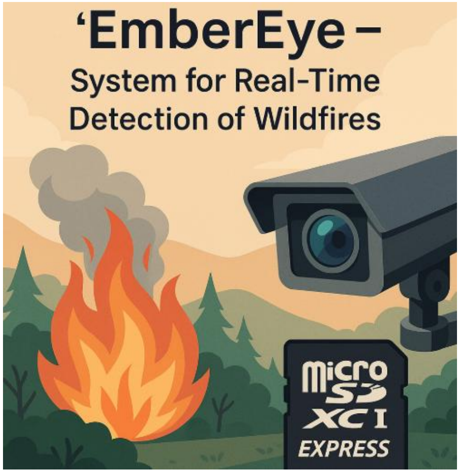


Abstract

Due to climate change, the world continues to experience wildfires that are often not detected in time. Many existing solutions are bulky, expensive, or impractical, leaving us with the growing problem of small, manageable fires turning into large-scale disasters.

Some of the approaches today are human watchtowers, which are labor-intensive and inefficient yet still in use, and satellites, which are weather-dependent and unable to detect small fires early enough.

Our approach is to use edge devices for scalable and affordable wildfire detection. They are more practical than human sentinels and operate closer to the ground than satellites, giving them the ability to spot small fires as they begin.

However, edge devices come with limitations: restricted RAM, modest CPU power, and slower memory. Running AI models on such hardware requires not only computational efficiency but also fast, sustained read/write performance to handle video streams and logging in real time.

In this project, we demonstrate how microSD Express storage, combined with carefully selected AI models, can overcome these limitations and enable reliable, real-time wildfire detection.

Introduction

The core challenge of wildfire detection is scale: vast areas need to be monitored, and detection must occur within minutes of ignition to allow a real chance of containment.

If we want to detect wildfires the moment they start, Cameras are a natural choice: they are affordable, scalable, and can directly capture smoke and flames in a way that can be verified by humans. However, using cameras introduces a trade-off between resolution and system requirements. Lower-resolution video eases the computational and storage load, but it also shrinks the area a single device can effectively monitor — meaning more cameras are needed to cover the same region. To overcome this, our system prioritizes **high-resolution cameras supported by fast storage**, ensuring wide-area detection without sacrificing detail or responsiveness. However, higher resolution puts additional strain on the hardware, especially on memory and storage throughput.

This is where **microSD Express** comes into play. With its high-speed read/write capabilities, our system can simultaneously run AI models for fire detection and record high-resolution video clips of fire events without delay. These clips are then transmitted to a human-operated station, where experts make the final decision on whether to dispatch firefighting services.

In summary, our complete system consists of a high-resolution camera, an NVIDIA Jetson Orin Nano, and microSD Express storage — together enabling scalable, affordable, and real-time wildfire detection at the edge.

System Overview

Our system is built as a fully containerized pipeline using Docker.

- An **LLM container** reads the camera stream and emits scene descriptions over UDP.
- A **classifier container** consumes these descriptions and outputs a FIRE/NO FIRE decision using a semantic text classifier.
- When three consecutive FIRE messages are detected, a **clip recorder** is triggered to save a high-resolution video segment into the mounted storage device via the LLM container.

The storage device under test (microSD Express) hosts the rolling video buffers, the evidence clips, and an 8 GB swap file used to sustain peak memory demand while inference runs (Figure 6).

Experiments & Results

We tested the system on recorded wildfire footage and a live camera feed. In both cases, the pipeline operated in real time, providing scene descriptions, fire/no-fire classification, event logging, and video clip saving without delay.

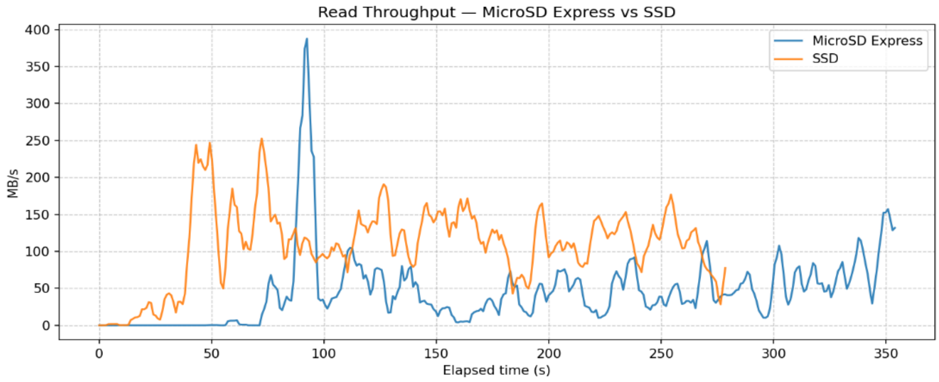
At the heart of the system sat a microSD Express card, which performed on par with a much larger SSD running the same system image. As shown in the Appendices, the throughput and CPU I/O wait graphs (Figures 1–5) reveal comparable behavior under continuous video capture and inference.

The Jetson Orin Nano relied on an 8 GB swap file (about 50% used in steady state), microSD Express maintained enough throughput to keep inference and video saving stable in real time. A conventional microSD card — typically limited to ~90 MB/s — would likely have been unable to support this load reliably, risking severe slowdowns or even process failure.

The detection model showed a low false-negative rate, consistently identifying visible fire events. High-resolution evidence clips were reliably saved for human verification and decision-making.

In summary, our experiments demonstrate that microSD Express can sustain demanding real-time workloads on edge devices, making it a practical enabler for compact and scalable wildfire detection systems.

Figure 1: the demonstration of the microSD Express impressive performance



Read Throughput — MicroSD Express vs SSD. Model running in real time on wildfire videos; only storage differs. Higher is better. Device read speed (MB/s). Peaks show bursts of frame loading. Higher is better for smooth video inference.

Discussion & Conclusion:

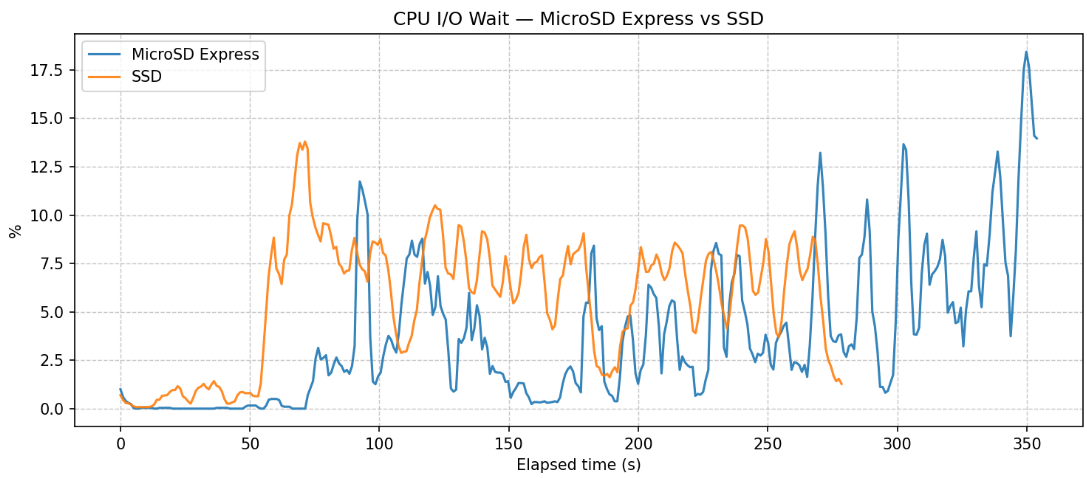
Our results show that microSD Express sustains real-time detection workloads without delay, removing storage as a bottleneck for edge devices. Its compact, low-cost form factor makes it practical for large-scale deployment, especially in remote regions where satellites and human watchtowers fall short.

The containerized design also ensures flexibility: new models or detection tasks — such as smoke, floods, or intrusions — can be added without changing the core pipeline. Overall, microSD Express demonstrates how fast, modern storage can unlock affordable and scalable edge AI solutions for wildfire detection and beyond.

Appendix

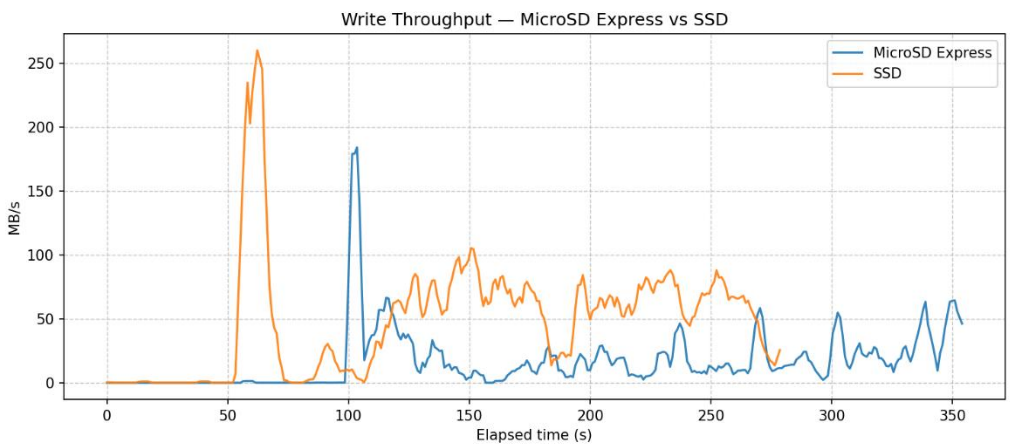
All of the graph figures present the operation of the model on an identical system image, once running from the microSD Express and once from the SSD. The peaks do not align perfectly because the models were started at different times, making it easier to compare their behavior. While the SSD shows slightly better performance in some cases, the microSD Express consistently delivered **comparable real-time performance under the exact same load**, proving it can keep up with much larger storage devices during this task.

Figure 2: again we can see really impressive performance of the microSD Express



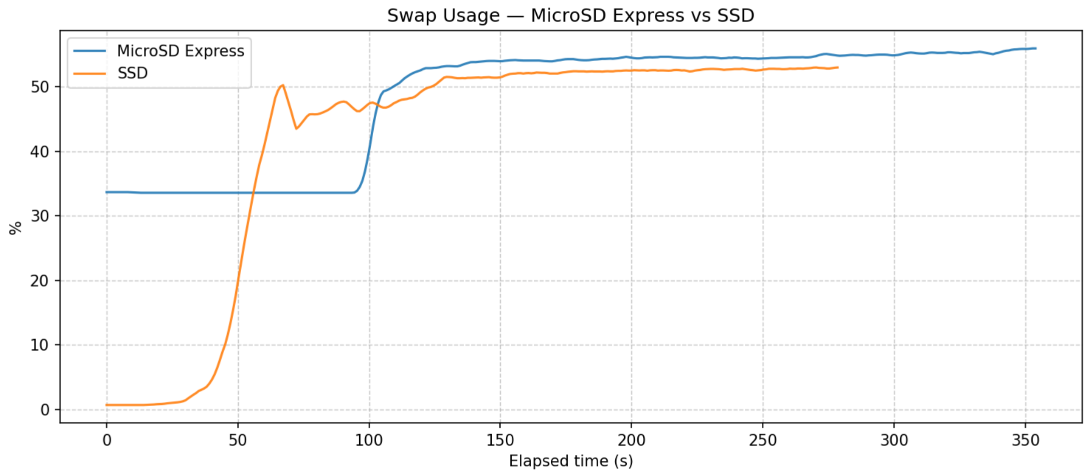
CPU I/O Wait — MicroSD Express vs SSD. Model running in real time on wildfire videos; only storage differs. Lower is better. CPU waiting for storage. Spikes = pipeline stalled by disk I/O.

Figure 3:



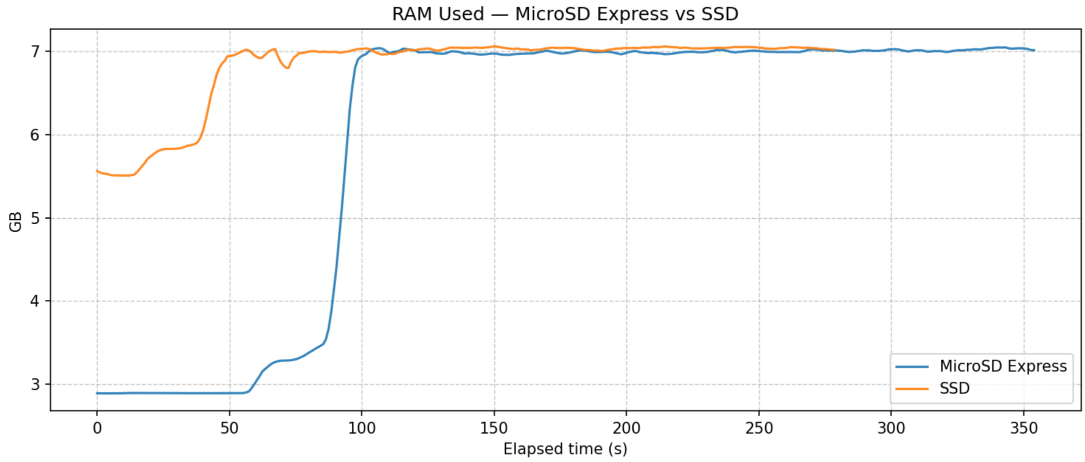
Write Throughput — MicroSD Express vs SSD. Model running in real time on wildfire videos; only storage differs. Higher is better. Clip write speed (MB/s). Peaks show bursts when saving outputs. Sustained low values may mean throttling.

Figure 4:



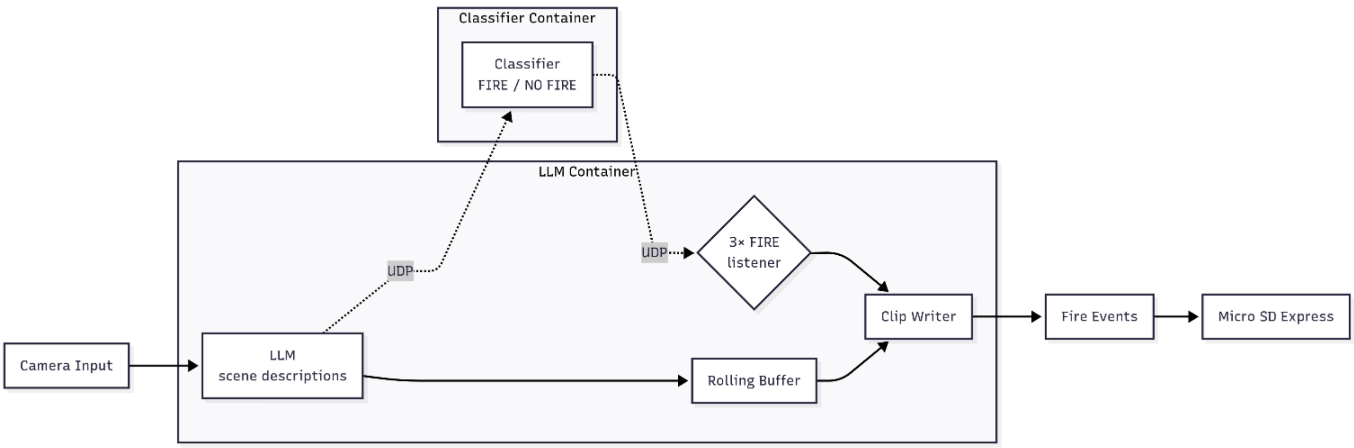
Swap Usage — MicroSD Express vs SSD. Model running in real time on wildfire videos; only storage differs. Swap use on disk. Growth means RAM is full, causing slowdowns.

Figure 5: we can see that we reach the maximum ram in the jetson – 7GB



RAM Used — MicroSD Express vs SSD (GB). Model running in real time on wildfire videos; identical pipeline, storage device is the only change. X-axis is elapsed seconds since the start of each run. MicroSD time range: 2025-09-15 17:53:46 → 2025-09-15 17:59:40. SSD time range: 2025-09-15 18:18:36 → 2025-09-15 18:23:15.

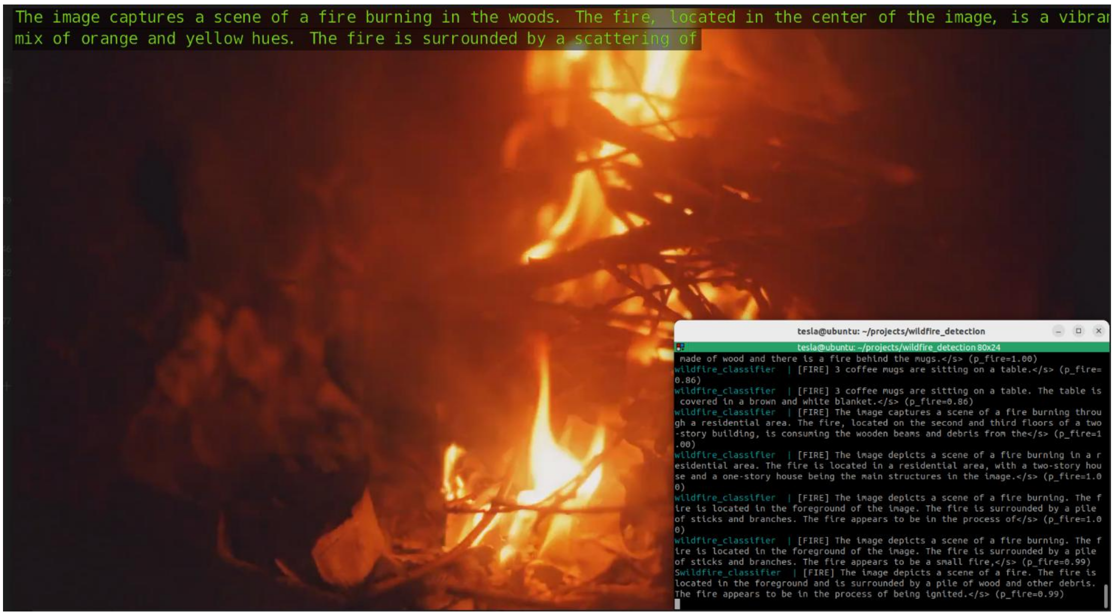
Figure 6: the block diagram of the project



Models used:

- Vision-Language Model: Efficient-Large-Model/VILA1.5-3b - chosen because it is a compact vision-language model that can generate accurate scene descriptions from video frames while still being lightweight enough to run on edge hardware.
- Text Classifier: typeform/distilbert-base-uncased-mnli - chosen because it is a fast, lightweight transformer model well-suited for semantic text classification, making it ideal for reliably deciding between FIRE and NO FIRE based on scene descriptions.

Figure 7: Screenshot of the wildfire detection pipeline in action. The video stream (left) is processed by the vision-language model to generate scene descriptions (top), which are then classified by the FIRE/NO FIRE model (bottom right).



Camera Specs: Video resolution of 1920×1080 at 30 FPS. The frame rate can be lowered to reduce processing load during model inference, since fire detection does not require 30 FPS. However, maintaining higher FPS is preferable for human verification of events.

SSD used: V-NAND SSD – 970EVOPlus NVMEM.2, 500 GB
Sequential Read: up to 3,500 MB/s
Sequential Write: up to 3,200 MB/s